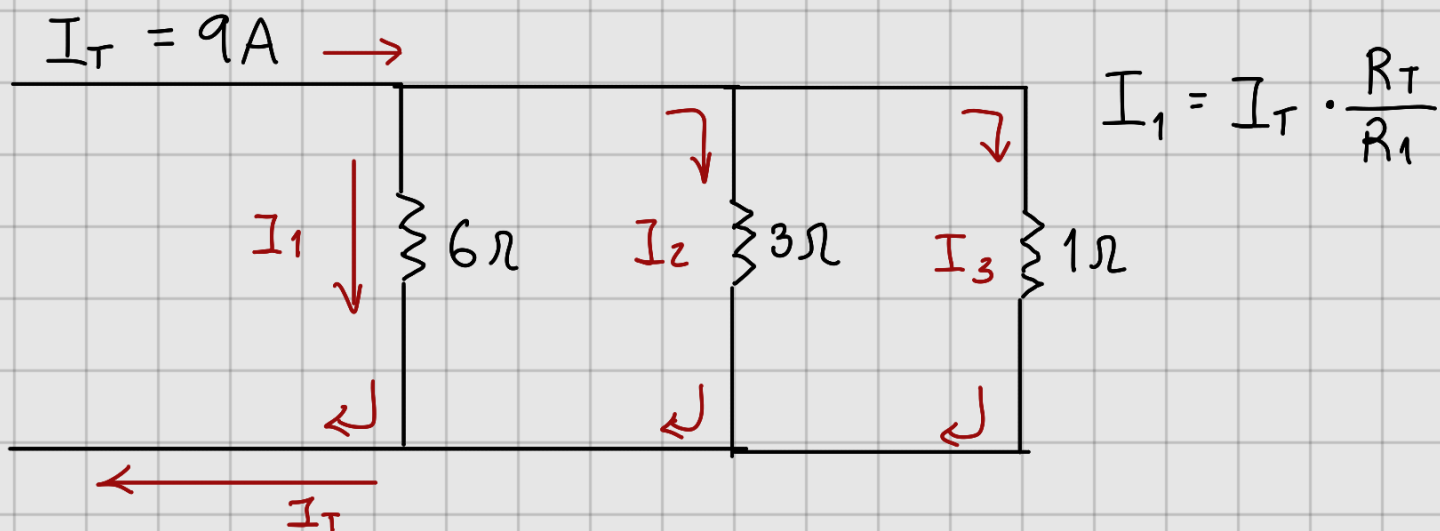




$$V_T = 6V \quad I_1 = 1A \quad I_2 = 2A \quad I_3 = 6A$$

$$R_T = 666m\Omega \quad \frac{1}{R_T} = 1.5S$$

$$R_T = \frac{1}{\frac{1}{6\Omega} + \frac{1}{3\Omega} + \frac{1}{1\Omega}} = \frac{1}{0.16\bar{6} + 0.3\bar{3} + 1} = \frac{1}{1.5\Omega}$$

$$= 0.66\bar{6}\Omega \quad V_T = R_T I_T = 666m\Omega(9A)$$

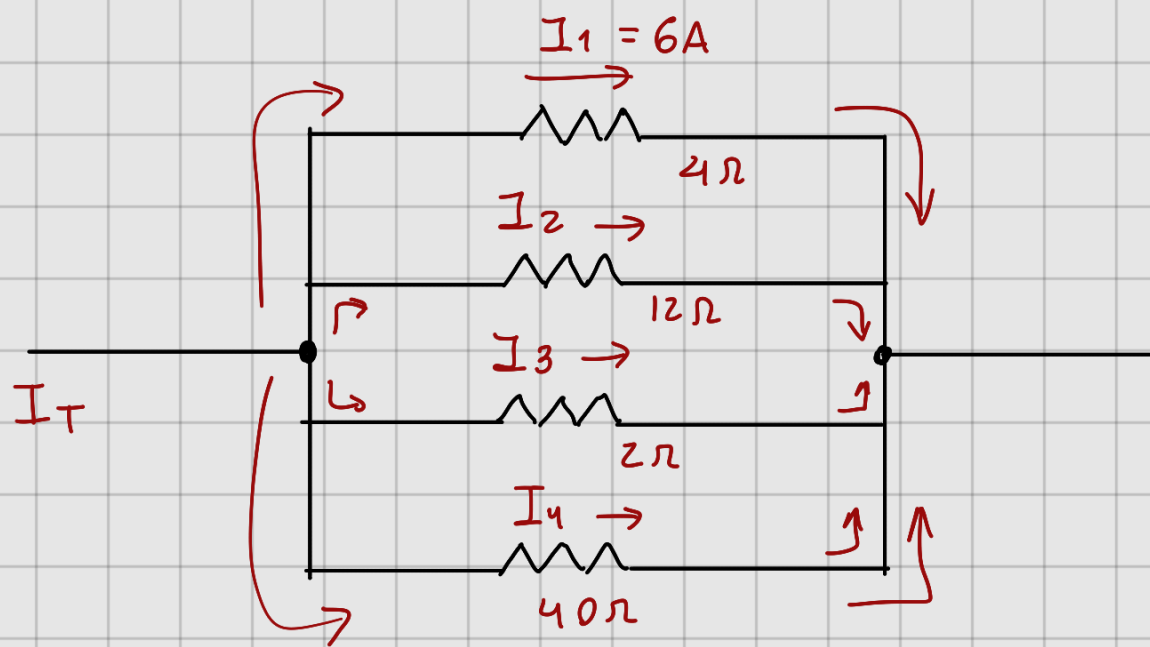
$$\underline{R_T = 666m\Omega} \quad \underline{= 6V}$$

$$I_1 = 9A \left(\frac{666m\Omega}{6\Omega} \right) = 1A$$

$$I_2 = 9A \left(\frac{666m\Omega}{3\Omega} \right) = 2A$$

$$I_3 = 9A \left(\frac{666m\Omega}{1\Omega} \right) = 6A$$

$$\frac{1}{R_T} = 1.5S$$



$$V = ? \quad I_1 = ? \quad I_2 = ? \quad I_3 = ? \quad I_4 = ?$$

$$I_T = ? \quad \frac{1}{R_T} = ?$$

$$I_1 = \frac{V_T}{R_1} \rightarrow V_T = I_1 R_1 = 6A(4\Omega) = 24V$$

$$R_T = \frac{1}{\frac{1}{4\Omega} + \frac{1}{12\Omega} + \frac{1}{2\Omega} + \frac{1}{40\Omega}} = 1.165\Omega$$

$$I_T = \frac{V_T}{R_T} = \frac{24V}{1.165\Omega} = 20.6A$$

$$I_2 = I_T \frac{R_T}{R_2} = 20.6A \left(\frac{1.165\Omega}{12\Omega} \right) = 2A$$

$$I_3 = I_T \frac{R_T}{R_3} = 20.6 A \left(\frac{1.165 \Omega}{2 \Omega} \right) = 12 A$$

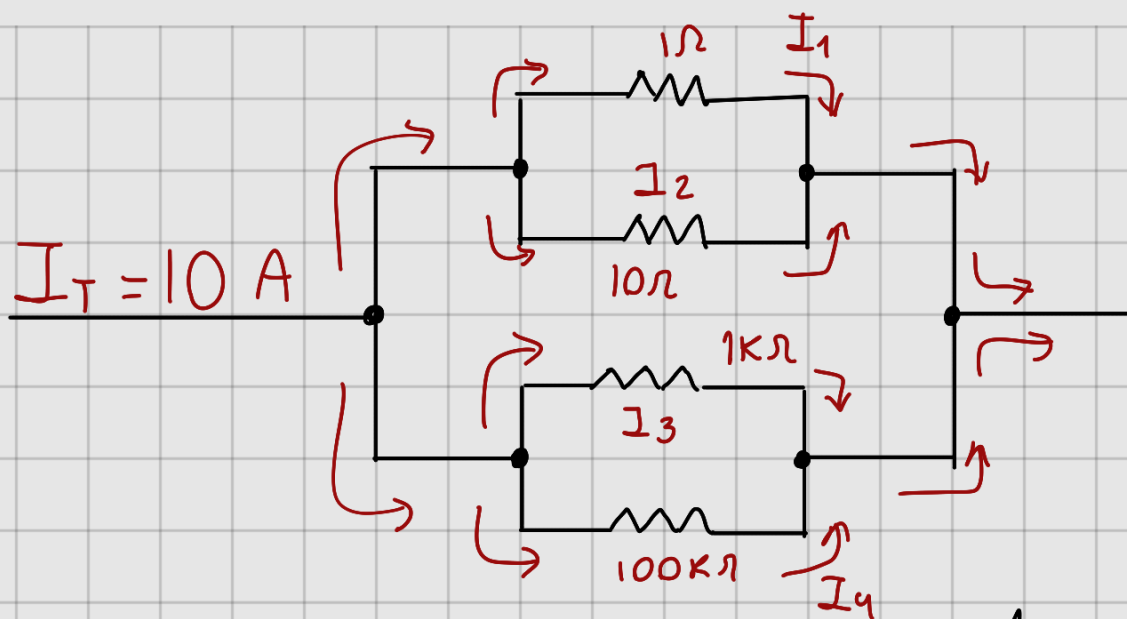
$$I_4 = I_T \frac{R_T}{R_4} = 20.6 A \left(\frac{1.165 \Omega}{40} \right) = 0.6 A$$

$$I_T = I_1 + I_2 + I_3 + I_4 = 6 A + 2 A + 12 A + 0.6 A \\ = 20.6 A$$

$$V_T = 24 V \quad I_1 = 6 A \quad I_2 = 2 A$$

$$I_3 = 12 A \quad I_4 = 0.6 A \quad I_T = 20.6 A$$

$$\frac{1}{R_T} = 0.85 S$$



$$I_T = 10\text{ A}$$

$$\frac{1}{R_T} = 1.11\text{ S}$$

$$R_T = \frac{1}{\frac{1}{1\Omega} + \frac{1}{10\Omega} + \frac{1}{1\text{ k}\Omega} + \frac{1}{100\text{ k}\Omega}} = 0.9\Omega$$

$$V_T = I_T R_T = 9\text{ V}$$

$$I_1 = I_T \frac{R_T}{R_1} = 10\text{ A} \left(\frac{0.9\Omega}{1\Omega} \right) = 9\text{ A}$$

$$I_2 = I_T \frac{R_T}{R_2} = 10\text{ A} \left(\frac{0.9\Omega}{10\Omega} \right) = 0.9\text{ A}$$

$$I_3 = I_T \frac{R_T}{R_3} = 10\text{ A} \left(\frac{0.9\Omega}{1000\Omega} \right) = 9\text{ mA}$$

$$I_4 = I_T \frac{R_T}{R_4} = 10\text{ A} \left(\frac{0.9\Omega}{100000\Omega} \right) = 90\text{ }\mu\text{A}$$

$$I_T = 10\text{ A} \quad R_T = 0.9\Omega \quad I_1 = 9\text{ A} \quad I_2 = 0.9\text{ A}$$

$$I_3 = 9\text{ mA} \quad I_4 = 90\text{ }\mu\text{A} \quad \frac{1}{R_T} = 1.11\text{ S}$$